University of Mascara
Department of Computer sciences
Second Year Engineer

Algorithmic and Complexity
Duration :1h30mn
January 8th, 2025

Final Exam

Exercise 1(*Sorting algorithms and complexities*):

1. Given the array [7, 2, 5, 3, 8, 1]:
- Compare **Selection Sort** and **Merge Sort** in terms of comparisons.
- Compare **Bubble Sort** and **Quick Sort** in terms of swaps.

2. Write a C function to determine if a given array contains any duplicate elements by using nested loops to compare every pair of elements in the array (naïve approach).
- Next, propose an optimized solution(C function) by first sorting the array using one of the methods presented in the lecture.
- Does sorting improve efficiency compared to the naive approach? why

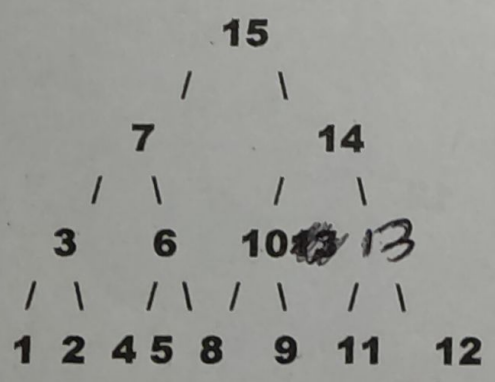
Exercise 2 (General tree)

Use the general tree structure presented in the lecture to represent family trees containing only male members. Each node represents a male person, and the children are linked as sons. Each person has a unique ID (integer). Implement C functions to

1. Check if a given person exists in the family tree.
2. Calculate the number of cousins for a given person.

Exercise 3 (Binary tree)

Write a function `create(int level)` to build a binary tree with *level*as given number of levels. The tree's nodes should have values starting from the leftmost leaf. Below is an example of a tree with 4 levels



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